

HARMAIN ALI BUTT

Santa Clara, CA | harmain.dev@gmail.com | +1 (650) 555-1234 | <https://www.linkedin.com/in/harmain-ali> | <https://github.com/Harmain-Ali>

Professional Summary

Highly accomplished and results-driven Senior Machine Learning Engineer with 8+ years of experience in developing, deploying, and optimizing Artificial Intelligence (AI) and Machine Learning (ML) models in production environments. Proven expertise in ML/AI tool development, robust quality control, and automating data analytics pipelines. Proficient in Python, C++, and JavaScript, with deep understanding of data science principles, linear algebra, algorithms, and statistics. Adept at leveraging frameworks like PyTorch and TensorFlow for scalable solutions and experienced in web development using Flask and Angular for interactive ML applications. Committed to monitoring output data quality in production environments to ensure high performance and reliability.

Skills

PROGRAMMING LANGUAGES:	Python, C++, JavaScript, SQL
MACHINE LEARNING & AI FRAMEWORKS:	PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Matplotlib, Plotly
WEB DEVELOPMENT & DATABASES:	Flask, Angular, ASP.Net Core, PostgreSQL, MongoDB (noSQL), RESTful APIs
CORE CONCEPTS:	Artificial Intelligence (AI), Machine Learning (ML), Data Science, Linear Algebra, Algorithms, Statistics, Quality Control, Model Deployment, Data Analytics Automation
TOOLS & PLATFORMS:	Git, Jupyter Notebooks, VS Code, Google Colab, Docker, AWS

Professional Experience

AMD | Santa Clara, CA Jan 2024 - Present

Senior Machine Learning Engineer

Technologies: Python, PyTorch, TensorFlow, C++, AWS, Docker, SQL, noSQL

- Led the development and deployment of advanced Artificial Intelligence (AI) models for optimizing GPU performance, resulting in a 15% improvement in processing efficiency across key product lines.
- Implemented robust quality control protocols and established automated monitoring systems to meticulously track output data quality in production environments, reducing critical errors by 20%.
- Architected and developed scalable ML/AI tool development pipelines using Python and C++, integrating complex algorithms and linear algebra concepts to enhance model training and inference capabilities.
- Collaborated with cross-functional teams to automate data analytics processes, identifying critical opportunities for ML and AI applications that drove strategic product enhancements.

NVIDIA | Santa Clara, CA Jan 2021 - Dec 2023

Machine Learning Engineer

Technologies: Python, TensorFlow, NumPy, Pandas, JavaScript, Flask, PostgreSQL

- Developed and deployed machine learning models for real-time image recognition and natural language processing, significantly improving data processing speeds by 25% using TensorFlow and Python.

- Contributed to ML/AI tool development, creating custom libraries that integrated advanced statistics and algorithms for feature engineering and model evaluation.
- Designed and implemented interactive web development interfaces for ML model visualization and interaction using JavaScript and Flask, enhancing user accessibility and data interpretation.
- Established rigorous quality control measures for deployed models, including continuous monitoring of output data quality in production environments to ensure model integrity and performance.

Google | Mountain View, CA Jan 2018 - Dec 2020

AI Developer

Technologies: Python, Scikit-learn, C++, SQL, Angular, ASP.Net Core

- Developed core Artificial Intelligence (AI) components for Google Cloud services, focusing on predictive analytics and anomaly detection using Python and C++.
- Engineered and deployed machine learning models, applying principles of linear algebra and statistics to optimize data-driven decision-making processes.
- Collaborated on web development projects, building user interfaces with Angular and ASP.Net Core to expose ML model functionalities and insights to internal stakeholders.
- Participated in the design of data pipelines for automating data analytics, ensuring data quality and consistency for downstream ML applications.

Key Projects

Advanced Stock Analysis Platform

Technologies: Yfinance, Python, Pandas, Plotly, Streamlit, Scikit-Learn, TensorFlow, SQL

- Architected and deployed a full-stack stock analysis platform that ingests real-time market data, computes 50+ fundamental and technical metrics, and generates investment scores with buy/hold/sell recommendations using Machine Learning models.
- Implemented a natural language summary generator that converts numeric scores into actionable insights, including strengths, risks, and contradictions, demonstrating advanced AI capabilities.
- Developed an interactive Streamlit dashboard with real-time ticker search, Plotly line charts, and modern gauge visualizations, ensuring high quality control over data presentation and user experience.

Automated Data Insights Engine

Technologies: Pandas, Numpy, Matplotlib, Plotly, Jupyter notebook, Python, Algorithms, Statistics

- Designed and implemented an automated data analytics engine for cleaning, visualizing, and analyzing complex datasets across multiple domains, extracting actionable insights using Python and advanced algorithms.
- Utilized comprehensive data science methodologies to identify patterns and trends through clear, interactive visualizations, enhancing data storytelling capabilities.
- Developed custom scripts to monitor output data quality, ensuring accuracy and reliability of insights generated from diverse datasets.

Certifications

- Machine Learning Engineering with TensorFlow on Google Cloud | Google Cloud
- Deep Learning Specialization | DeepLearning.AI (Andrew Ng)
- Professional Certificate in Python for Data Science and AI | IBM
- Advanced JavaScript and Front-End Development | Udemy

Education

Stanford University

Master's in Computer Science, specializing in Artificial Intelligence Sep 2016 - Dec 2017 **University of California, Berkeley**

Bachelor of Science in Computer Engineering Sep 2012 - May 2016